Generating Images in Few Steps:
Flow Matching versus DDPM

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https://github.com/xHansi/flow-matching-fewstep

Abstract

Diffusion models produce excellent images but they need many sampling steps, which makes generation slow. Flow Matching learns a nearly straight velocity field whose trajectory can be followed in only a handful of steps. We implement a Flow Matching model and a DDPM baseline on top of one shared network and we ask how image quality measured by FID changes as we vary the number of sampling steps. On MNIST the Flow Matching model already returns clean digits at two steps while the baseline still returns noise. It reaches near best quality around four to eight steps. The advantage carries over to the harder Fashion MNIST domain even though the gap becomes smaller.

1. Introduction

Image generation today is dominated by diffusion. A diffusion model such as DDPM (Ho et al., 2020) learns to reverse a slow noising process and it draws an image by walking backward through many small denoising steps. The quality is impressive but the cost is real. A single image can require dozens or even hundreds of network evaluations, which becomes a problem whenever generation has to be fast or cheap. This raises a natural question. Do we truly need that many steps or is the long trajectory an artifact of the particular formulation?

Flow Matching (Lipman et al., 2023) and the closely related Rectified Flow (Liu et al., 2023) offer a different view. Instead of a stochastic denoising chain they learn a velocity field that carries noise to data along paths that stay close to straight lines. A straight path can be followed with very few steps of a simple solver, so the hope is to keep the quality of diffusion while cutting the number of evaluations by an order of magnitude. In this project we test that hope directly. We build a Flow Matching model and a DDPM baseline that share one network and the same training budget, we sample from both across a wide range of step counts and we measure FID at every setting. We run the experiment on MNIST first and then on Fashion MNIST, so we can see whether the conclusion depends on the dataset.

2. Related work

Denoising diffusion probabilistic models were introduced in (Ho et al., 2020) and made faster by the deterministic DDIM update (Song et al., 2021). Flow Matching (Lipman et al., 2023) and Rectified Flow (Liu et al., 2023) instead learn a continuous velocity field. Rectified Flow adds the reflow step used here. We rely on classifier free guidance (Ho & Salimans, 2021) and on the Fréchet Inception Distance (Heusel et al., 2017) over InceptionV3 features (Szegedy et al., 2016). The datasets are MNIST (LeCun et al., 1998) and Fashion MNIST (Xiao et al., 2017).

3. Method

Shared network. Both methods use one convolutional network of about six and a half million parameters that is conditioned on the diffusion time. The class label enters through a learned embedding that carries one extra null token, which lets us apply classifier free guidance at sampling time. We sample from an exponential moving average of the weights.

Flow Matching. We draw a noise sample x_0 and a data sample x_1 and interpolate linearly as $x_t = (1-t)\,x_0 + t\,x_1$. The network is trained to predict the constant velocity $x_1 - x_0$ under a mean squared error loss. To generate an image we start from noise and integrate this velocity with an Euler solver. The number of Euler steps is exactly the quantity we vary in the experiments.

DDPM baseline. The baseline follows the standard recipe with a linear noise schedule and a network that pre-

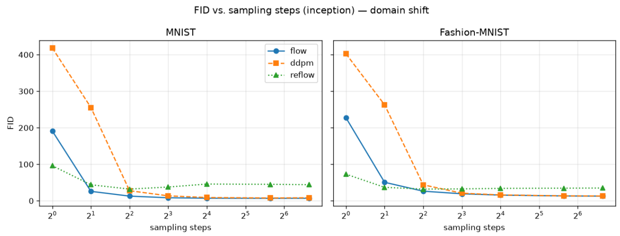

Figure 1. FID against the number of sampling steps on MNIST (left) and Fashion MNIST (right). Flow Matching is far ahead of the diffusion baseline in the few step regime on both datasets and the two methods converge once many steps are used. Lower is better.

dicts the added noise. For sampling we use the DDIM update, which is deterministic and lets us respace the schedule to any number of steps.

Reflow. Starting from the trained Flow Matching model we generate pairs that link a noise sample to the image it produces and retrain a fresh model on them. Because the pairs are already coupled the learned trajectories become straighter, which should help at the very low end of the step budget.

Evaluation. We report FID with InceptionV3 features as the standard choice and also with features from a small classifier trained on the same dataset, which fits grayscale images better. FID is measured without guidance because guidance inflates the score and interacts with the step count.

4. Results

Few steps. Figure 1 is the central result. On MNIST the Flow Matching model reaches an Inception FID near eight already at eight steps while the diffusion baseline needs many more to get there. At two steps the difference is dramatic. Flow Matching scores about twenty six against about two hundred fifty five for the baseline, a factor of ten. Table 1 lists the numbers at selected step counts. Reflow does even better at the very low end. After a heavier reflow stage it reaches an Inception FID of ten at two steps, below plain Flow Matching, because its trajectories are straighter. The appendix shows the qualitative side, a Flow Matching sample turning from noise into a digit over eight steps (Figure 2) together with a two step comparison of both methods on both datasets (Figure 3).

Guidance and step count. With strong guidance the diffusion baseline degrades at high step counts and the samples oversaturate, because deterministic guidance compounds over a long trajectory. Without guidance the same

Table 1. Inception FID at selected step counts on MNIST, measured without guidance. Lower is better.

steps	2	4	8	50
Flow Matching	25.9	12.9	**8.4**	**6.9**
DDPM	255.5	27.6	13.7	7.8
Reflow	**10.0**	**8.7**	**8.4**	8.1

model is clean again at one hundred steps. This is why we exclude guidance from the FID measurement.

Domain shift. When we move from digits to clothing on Fashion MNIST the story holds. Flow Matching keeps its lead in the few step regime and both methods converge once many steps are used. Two things change. The absolute FID roughly doubles because clothing is more textured than digits. The baseline also catches up a little earlier. At eight steps the two are almost tied on Fashion MNIST while Flow Matching was still clearly ahead on MNIST. The direction of the result is the same but its size shrinks on the harder domain.

5. Discussion and conclusions

Our experiments support the promise of Flow Matching. With a shared network and an equal training budget it produces usable images in one to four steps where the diffusion baseline still needs many more. The advantage also survives a change of dataset. Reflow gives the biggest gain in the very low step regime. With enough reflow pairs its straighter paths reach the quality of many step diffusion in only two steps, which is the strongest single piece of evidence in our study for the few step promise. A practical lesson is that the guidance scale must be fixed when comparing samplers, because it interacts strongly with the number of steps. The main limitation of our study is scale. We work at low resolution on two small datasets and a larger benchmark would make the conclusions stronger.

6. Statement on the use of AI

AI based tools were used as a support during this project. I used chat based assistants, mainly ChatGPT and Google Gemini, for boilerplate code, for looking up library details, for debugging and for polishing the writing of this report. The research question and the hypothesis, the choice of models and metrics, the design of the domain shift experiment and the reading of the numbers are my own. Every figure and every FID value comes from an actual training run and I verified the outputs myself before drawing any conclusion. AI was not used to invent or embellish any result.

References

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Appendix

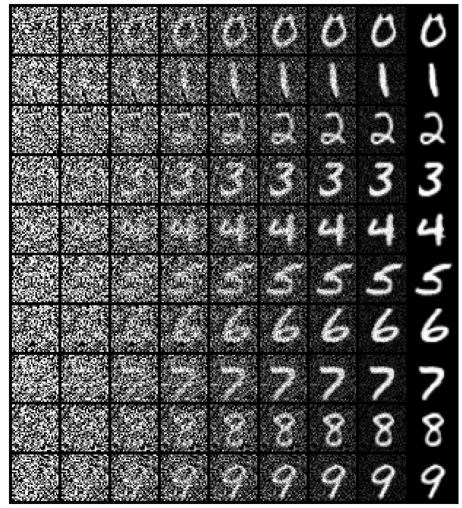

Figure 2. Sampling trajectory of the Flow Matching model. Each row is one digit and the columns go from pure noise on the left to the final image on the right over eight Euler steps. The class is legible after one or two steps.

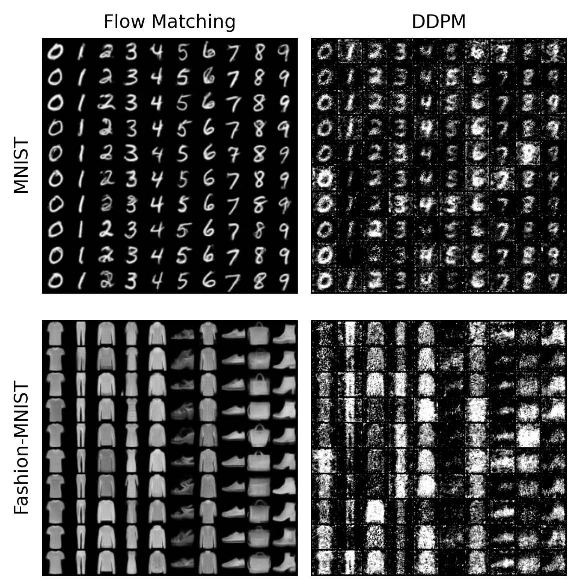

Figure 3. Samples at two sampling steps, without guidance to match the FID setting. Flow Matching (left) already produces recognizable digits and clothing, while the DDPM baseline (right) is still essentially noise, on both MNIST (top) and Fashion MNIST (bottom).